

820, that means all the sensors are straight, i.e., the wheelchair is not moving. If any sensor value exceeds this threshold limit, the corresponding motor starts rotating. Simultaneously, a message is displayed on LCD1 along with indication on the corresponding LED.

While testing at EFY Lab, PROGISP programmer was used to program the IC.

Construction and testing

An actual-size PCB layout for ATmega32-based gesture control of a wheelchair is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit on the PCB, enclose it in a suitable box.

Fix CON1 and CON2 on the back side of the cabinet for connecting 5V and 12V power supplies, respectively. Affix switch S1 (used for resetting the circuit) on the front side. Attach four sensors (FS1 through FS4) to your four fingers such that you can bend all the fingers easily

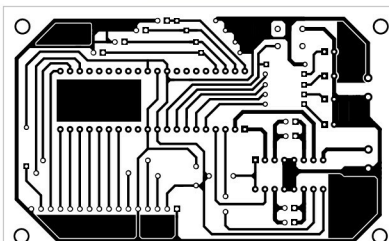


Fig. 5: Actual-size PCB layout for gesture control of a wheelchair

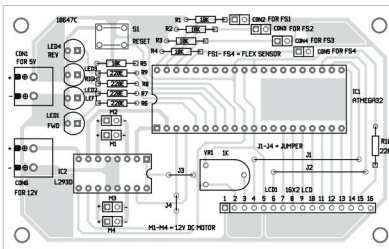


Fig. 6: Components layout for the PCB

along with sensors.

Operation. Initially, LCD1 displays the message "Gesture control wheel chair." Then it displays the direction of motor movements. If the wheelchair is not moving, LCD1 displays the message "Wheel chair is stopped."

When all the fingers are straight, it means all the sensors are straight. The output from a straight sensor is less than the threshold limit, so the microcontroller does not give signals to any of the motors. So the wheelchair does not move.

First finger. Now, if you bend

your first finger (index finger) with FS1 sensor attached to it, the output from FS1 exceeds the threshold limit. The microcontroller reads its value and rotates all the four motors forward. So the wheelchair moves in forward direction. LCD1 displays the message "wheel chair is moving forward" and LED1 glows to indicate that the wheelchair is moving forward.

Second finger. When you bend your second finger (middle finger), the output from FS2 exceeds the threshold limit. The microcontroller reads its value and rotates both the right-side motors forward. So the wheelchair turns left. LCD1 displays the message "wheel chair is turning left" and LED2 glows to indicate that the wheelchair is turning left.

Third finger. Similarly, when you bend your third finger, the output from FS3 exceeds the threshold limit. This time the microcontroller rotates both the left motors forward. So the wheelchair turns right. LCD1 displays the message "wheel chair is turning right" and LED3 glows to indicate that the wheelchair is turning right.

Fourth finger. Finally, when you bend your fourth finger, the output from FS4 exceeds the threshold limit. The microcontroller rotates all the four motors in reverse direction. LCD1 displays the message "wheel chair is moving reverse" and LED4 glows to indicate that the wheelchair is moving in reverse direction. **EFY**

PARTS LIST

Semiconductors:

- IC1 - ATmega32 microcontroller
- IC2 - L293D motor driver

LED1-LED4

- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1-R5 - 10-kilo-ohm
- R6-R10 - 220-ohm
- VR1 - 1-kilo-ohm preset

Miscellaneous:

- CON1, CON6 - 2-pin terminal connector
- CON2-CON5 - 2-pin connector
- LCD1 - 16x2 line LCD
- MI-M4 - 12V DC motor
- S1 - Tactile switch
- FS1-FS4 - Flex sensor
- 5V regulated power supply
- 12V battery/regulated power supply

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



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